

What is National Income? Concept, definition, measurement.

What is National Income?

National Income of any country means the complete value of the goods and services produced by any country during its financial year. It is, thus the consequence of all economic activities that are running in any country during the period of one year. It is valued in terms of money. In short one can say that the national income of any country is the total amount of income that is accrued by it through various economic activities in one year. It is also helpful in determining the progress of the country. It includes wages, interest, rent, profit, received by factors of production like labour, capital, land and entrepreneurship of a nation.

National Income is defined as the net value of all economic goods and services produced within the domestic territory of a country in an accounting year plus the net factor income from abroad. According to the national bureau of statistics (NBS), 'National income is the sum total of factor incomes generated by the normal residents of a country in the form of wages, rent, interest and profit in an accounting year'.

Different Concepts of National Income

National income accounts have three sides: a product side, an expenditure side and an income side. The product side measures production based on concept of value added. The expenditure side looks at the final sales of goods and services. Whereas the income side measures the distribution of the proceeds from sales to different factors of production. Accordingly, national income is a measure of the total flow of 'earnings of the factor-owners' which they receive through the production of goods and services. Thus, national income is the sum total of all the incomes accruing over a specified period to the residents of a country and consists of wages, salaries, profits, rent and interest.

There are various concepts of National Income including GDP, GNP, NNP, NI, PI, DI, and PCI which explain the facts of economic activities.

a. GDP at market price: Gross domestic product (GDP) is a measure of the market value of all final economic goods and services, gross of depreciation, produced within the domestic territory of a country during a given time period. Is money value of all goods and services produced within the domestic domain with the available resources during a year.

$$GDP = (P \times Q)$$

Where,

GDP = gross domestic product

P = Price of goods and services

Q = Quantity of goods and services

GDP is made up of 4 Components

- a. consumption
- b. investment
- c. government expenditure
- d. net foreign exports of a country

$$\text{GDP} = C + I + G + (X - M)$$

Where,

C = Consumption

I = Investment

G = Government expenditure

(X - M) = Export minus import

b. Gross National Product (GNP)

Gross National Product (GNP) is a measure of the market value of all final economic goods and services, gross of depreciation, produced within the domestic territory of a country by normal residents during an accounting year including net factor incomes from abroad. Gross National Product (GNP) is evaluated at market prices and therefore it is in fact Gross

$$\text{GNP}_{\text{MP}} = \text{GDP}_{\text{MP}} + \text{Net Factor Income from Abroad}$$

National Product at market prices (GNP_{MP}).

$$\text{GDP}_{\text{MP}} = \text{GNP}_{\text{MP}} - \text{Net Factor Income from Abroad}$$

NFIA is the difference between the aggregate amount that a country's citizens and companies earn abroad, and the aggregate amount that foreign citizens and overseas companies earn in that country.

If Net Factor Income from Abroad is positive, then GNP_{MP} would be greater than GDP_{MP} .

Is market value of final goods and services produced in a year by the residents of the country within the domestic territory as well as abroad. GNP is the value of goods and services that the country's citizens produce regardless of their location.

$$\text{GNP} = \text{GDP} + \text{NFIA} \text{ or,}$$

$$\text{GNP} = C + I + G + (X - M) + \text{NFIA}$$

Where,

C = Consumption

I = Investment

G = Government expenditure

(X - M) = Export minus import

NFIA = Net factor income from abroad.

c. Net National Product at Market Prices (NNP_{MP})

Net National Product at Market Prices (NNP_{MP}) is a measure of the market value of all final economic goods and services, produced by normal residents within the domestic territory of a country including Net Factor Income from Abroad during an accounting year excluding depreciation.

$$NNP_{MP} = GNP_{MP} - \text{Depreciation}$$

$$NNP_{MP} = NDP_{MP} + \text{Net Factor Income from Abroad}$$

$$NNP_{MP} = GDP_{MP} + \text{Net Factor Income from Abroad} - \text{Depreciation}$$

Is market value of net output of final goods and services produced by an economy during a year and net factor income from abroad.

$$NNP = GNP - \text{Depreciation}$$

$$\text{or, } NNP = C + I + G + (X - M) + NFIA - IT - \text{Depreciation}$$

Where,

C = Consumption

I = Investment

G = Government expenditure

(X - M) = Export minus import

NFIA = Net factor income from abroad.

IT = Indirect Taxes

d. National Income (NI): Is also known as National Income at factor cost which means total income earned by resources for their contribution of land, labour, capital and organisational ability. Hence, the sum of the income received by factors of production in the form of rent, wages, interest and profit is called National Income.

Symbolically or as per the formula

$$NI = NNP + \text{Subsidies} - \text{Interest Taxes}$$

$$\text{or, } GNP - \text{Depreciation} + \text{Subsidies} - \text{Indirect Taxes}$$

$$\text{or, } NI = C + G + I + (X - M) + NFIA - \text{Depreciation} - \text{Indirect Taxes} + \text{Subsidies}$$

e. Personal Income (PI): Is the total money income received by individuals and households of a country from all possible sources before direct taxes.

Personal Income While national income is income earned by factors of production, Personal Income is the income received by the household sector including Non-Profit Institutions Serving Households. Thus, national income is a measure of income earned and personal income is a measure of actual current income receipts of persons from all sources which may or may not be earned from productive activities during a given period of time. In other words, it is the income 'actually paid out' to the household sector, but not necessarily earned. Examples of this include transfer payments such as social security benefits, unemployment compensation, welfare payments etc. Individuals also contribute income which they do not actually receive, for example, undistributed corporate profits and the contribution of

$$PI = NI + \text{Income received but not earned} - \text{Income earned but not}$$

employers to social security. Personal income forms the basis for consumption expenditure and is derived from national income as follows:

An important point to remember is that national income is not the sum of personal incomes because personal income includes transfer payments (eg. pension) which are excluded from national income. Also, not all national income accrues to individuals as their personal income. Therefore, personal income can be expressed as follows:
$$PI = NI - \text{Corporate Income Taxes} - \text{Undistributed Corporate Profits} - \text{Social Security Contribution} + \text{Transfer Payments}.$$

f. Disposable Personal Income (DI)

Disposable personal income is a measure of amount of the money in the hands of the individuals that is available for their consumption or savings. Disposable personal income is derived from personal income by subtracting the direct taxes paid by individuals and other compulsory

$$DI = PI - \text{Personal Income Taxes}$$

payments made to the government. It is the income left with the individuals after the payment of direct taxes from personal income. It is the actual income left for disposal or that can be spent for consumption by individuals.

Thus, it can be expressed as:

$$DI = PI - \text{Direct Taxes}$$

Per Capita Income

The GDP per capita is a measure of a country's economic output per person. It is obtained by dividing the country's gross domestic product, adjusted by inflation, by the total population. It serves as an indicator of the standard of living of a country. It is calculated by dividing the national income of the country by the total population of a country. Thus, $PCI = \text{Total National Income} / \text{Total National Population}$

Measurement of National Income

There are three methods to calculate National Income:

1. Income Method
 2. Product/ Value Added Method
 3. Expenditure Method
- Income Method

In this National Income is measured as flow of income.

We can calculate NI as:

Net National Income = Compensation of Employees + Operating surplus mixed (w + R + P + I) + Net income + Net factor income from abroad.

Where,

W = Wages and salaries

R = Rental Income

P = Profit

I = Mixed Income

- **Product/ Value Added Method**

In this National Income is measured as flow of goods and services.

We can calculate NI as:

NATIONAL INCOME = G.N.P - COST OF CAPITAL - DEPRECIATION - INDIRECT TAXES

- **Expenditure Method**

In this National Income is measured as flow of expenditure.

We can calculate NI through Expenditure method as:

National Income = National Product = National Expenditure.

EFFECTS OF DEDUCTIONS ON PERSONAL INCOMES:

A tax deduction reduces your Adjusted Gross Income or AGI on your income tax return, thus either increasing your tax refund or reducing your taxes. It's not just about how much income you make, but how much you get to keep of your own pie.

There is a large body of evidence that shows that high state PIT rates have a negative effect on business and individual decisions and thus slow the growth of employment and per capita income in a state. At the individual level, personal income taxes influence work effort and state-to-state migration. At the firm level, the evidence suggests that state PIT rates influence business location decisions and entrepreneurs' decisions to grow their firms by hiring

employees and investing in plant, facilities, and equipment. Thus, decreasing PIT rates, particularly if they are significantly higher than those in neighbouring and other competitor states, leads to growth in employment and per capita income. Economic growth, in turn, helps to reduce the poverty rate.

In short, it does not require a 'leap of faith' to believe that the benefits of decreasing the PIT rate on high income individuals are broadly shared throughout the income distribution.

Indeed, the evidence shows that states with lower PIT rates enjoy higher growth in employment and real per capita income.

USEFULNESS AND SIGNIFICANCE OF NATIONAL INCOME ESTIMATES

1. National income accounts provide a comprehensive, conceptual and accounting framework for analyzing and evaluating the short-run performance of an economy. The level of national income indicates the level of economic activity and economic development as well as aggregate demand for goods and services of a country.
2. The distribution pattern of national income determines the pattern of demand for goods and services and enables businesses to forecast the future demand for their products.
3. Economic welfare depends to a considerable degree on the magnitude and distribution of national income, size of per capita income and the growth of these over time.
4. The estimates of national income show the composition and structure of national income in terms of different sectors of the economy, the periodical variations in them and the broad sectoral shifts in an economy over time. It is also possible to make temporal and spatial comparisons of the trend and speed of economic progress and development. Using these information, the governments can fix various sector-specific development targets for different sectors of the economy and formulate suitable development plans and policies to increase growth rates.
5. National income statistics also provide a quantitative basis for macroeconomic modeling and analysis, for assessing and choosing economic policies and for objective statement as well as evaluation of governments' economic policies. These figures often influence popular and political judgments about the relative success of economic programmes.
6. National income estimates throw light on income distribution and the possible inequality in the distribution among different categories of income earners. It is also possible to make comparisons of structural statistics, such as ratios of investment, taxes, or government expenditures to GDP.
7. International comparisons in respect of incomes and living standards assist in determining eligibility for loans, and other funds or conditions on which such loans, and/or funds are made available. The national income data are also useful to determine the share of nation's contributions to various international bodies.
8. Combined with financial and monetary data, national income data provide a guide to make policies for growth and inflation.

9. National income or a relevant component of it is an indispensable variable considered in economic forecasting and to make projections about the future development trends of the economy.

Formula to Calculate National Income

National Income Formula refers to the formula that is used in order to calculate value of total items manufactured in-country by its residents and income received by its residents and as per the formula, national income is calculated by adding together consumption, government expenditure, investments made within the country, its net exports i.e., exports minus imports, foreign production by resident of country and then subtracting the domestic production by residents of other country.

$$\text{National Income} = C + G + I + X + F - D$$

- C is the Consumption
- G is the government expenditure
- I is the Investments
- X is Net Exports (Exports less Imports)
- F is the National Resident's Foreign Production
- D is the Non-National Resident's Domestic Production

Step by Step Calculation Methods of National Income

The following are the methods to calculate national income using its formula.

1. The first part is the consumption that needs to be identified and computed and that is nothing, but total expenditure incurred by the country's government in the procurement of goods and services.
2. Infrastructure, capital investments, government employee salary shall form part of total investments made by the government.
3. Total investments made within the country also needs to figure out.
4. Calculate the export value of goods and services that have been produced within the country.
5. Imports Value needs to be calculated as well so this can be excluded for the calculation of national income.
6. Next, find out the value of national production by foreign residents.
7. Now figure out the value of foreign production by national residents.
8. Now sum up all the values from step 1 to step 4 and deduct values computed in step 5 and step 6 and lastly add value arrived at step 7.

Examples

We are given the following hypothetical inputs in Nigeria ₦ Billion for economy of country XYZ. You are required to calculate the national income of the country XYZ.

- Consumption (C): ₦ 10
- Government Expenditure (G): ₦ 14

- Investments (I): ~~£~~ 24
- Depreciation: ~~£~~ 2
- Exports: ~~£~~ 8
- Imports: ~~£~~ 4
- Net Exports (X): ~~£~~ 4
- Foreign Production by National Resident (F): ~~£~~ 1
- Domestic Production by Non-Resident (D): ~~£~~ 3

Solution:

Therefore, the calculation of the national income is as follows,

$$NI = C + G + I + F - D = \text{N} 10 + 14 + 24 + (8 - 4) + 1 - 3 = \text{N} 50;$$

Hence, the National Income of country XYZ is ~~£~~ 50

	National Income = C + G + I + X + F - D		= B2 + B3 + B4 + B8 + B9 - B10
1	Particulars	Amount	
2	Consumption (C)	\$10	
3	Government Expenditure (G)	\$14	
4	Investments (I)	\$24	
5	Depreciation	\$2	
6	Exports	\$8	
7	Imports	\$4	
8	Net Exports (X)	\$4	
9	Foreign Production by National Resident (F)	\$1	
10	Domestic Production by Non-Resident (D)	\$3	
11	National Income		= B2 + B3 + B4 + B8 + B9 - B10
12			

National Income will be = ~~£~~ 50 Billion

NB: Depreciation is not taken into consideration.

EXAMPLE 2

Country Y & Z are the two countries wherein the world bank was confused about which country to be rank over the other. The GDP of the two countries was ~~£~~ 6,000 Million approximately and therefore the bank decided to rank them on the basis of National Income. Following details were gathered:

Particulars	Country Y	Country Z
GDP	2000.00	2000.00
Government Expenditure (G)	600.00	700.00
Investments (I)	120.00	320.00
Foreign Production by National Resident (F)	100.00	200.00
Domestic Production by Non-Resident (D)	300.00	100.00

Based on the above information, you are required to do the calculation of National Income formula and rank which country would be superior to another?

Solution

In this example, we are not given all the inputs that are required to calculate National Income but there are certain inputs of National Income which if combined will form GDP which is nothing but summing up Consumption, Government Expenditure, Investments, Net Exports and we are given that and hence we shall use GDP as proxy for same and calculate the National Income.

Therefore, the calculation of the national income for country Y is as follows,

National Income = GDP + F - D		= B2 + B5 - B6	
1	Particulars	Country XYZ	
2	GDP	2000.00	
3	Government Expenditure (G)	600.00	
4	Investments (I)	120.00	
5	Foreign Production by National Resident (F)	100.00	
6	Domestic Production by Non-Resident (D)	300.00	
7	National Income	= B2 + B5 - B6	
8			

- = (C + G + I + X) + F - D
- = GDP + F - D
- = 2000.00 + 100.00 - 300.00

National Income for country Y & Z will be --

C7				=C2-C6+C5
A				
1	Particulars	Country (BnR)		
2	GDP		2000.00	
3	Government Expenditure (G)		700.00	
4	Investments (I)		320.00	
5	Foreign Production by National Resident (F)		200.00	
6	Domestic Production by Non-Resident (D)		100.00	
7	National Income		2100.00	
8				

* = 2,100

If the bank takes National Income as a decider to rank them then Country Z will be rank above Country XYZ as Country XYZ has National Income higher by \$300 billion.

DGG is considering investing in the country where the National Income of the country is a minimum of ₦1,300 billion. Below are the three developing nations which they have shortlisted and are considering investing in:

Particulars	Country A	Country B	Country C
GDP	2000	1500	1200
Government Expenditure (G)	2800	2100	1680
Investments (I)	4800	3600	2880
Exports	1600	1200	960
Imports	7900	7500	6100
Net Exports (X)	-6300	-6300	-5140
Foreign Production by National Resident (F)	200	150	120
Domestic Production by Non-Resident (D)	600	450	360

All three countries highly import oriented countries.

DGG is looking to invest \$500 million. Based on National Income, you are required to determine where would DGG invest in?

Solution

Calculation of the national income for country A is as follows,

National Income = $C + G + I + X + F - D$			Country / No		E
2	Resident				
3	Consumption (C)		2000		
4	Government Expenditure (G)		2800		
5	Investments (I)		4800		
6	Exports		1600		
7	Imports		7900		
8	Net Exports (X)		-6300		
9	Foreign Production by National Resident (F)		200		
10	Domestic Production by Non-Resident (D)		600		
11	National Income		$=B3+B4+B5+B8+B9-B10$		
12					

• $=2000+2800+4800+(-6300)+200-600$

National Income for country A will be -

B11					$=B3+B4+B5+B8+B9-B10$
1		A		B	E
2	Resident				
3	Consumption (C)		2000		
4	Government Expenditure (G)		2800		
5	Investments (I)		4800		
6	Exports		1600		
7	Imports		7900		
8	Net Exports (X)		-6300		
9	Foreign Production by National Resident (F)		200		
10	Domestic Production by Non-Resident (D)		600		
11	National Income		2900		
12					

• $=2900$

Similarly, we can calculate national income for country B & country O as shown below,

National Income for country N will be -

C11	A	C	E
	Country A	Country C	Country E
2	Population		
3	Consumption (C)	1500	
4	Government Expenditure (G)	2100	
5	Investments (I)	3600	
6	Exports	1200	
7	Imports	7500	
8	Net Exports (X)	-6300	
9	Foreign Production by National Resident (F)	150	
10	Domestic Production by Non-Resident (D)	450	
11	National Income	600	
12			

• =600

National Income for country C will be -

D11	A	D	E
	Country A	Country D	Country E
2	Population		
3	Consumption (C)	1200	
4	Government Expenditure (G)	1680	
5	Investments (I)	2880	
6	Exports	960	
7	Imports	6100	
8	Net Exports (X)	-5140	
9	Foreign Production by National Resident (F)	120	
10	Domestic Production by Non-Resident (D)	360	
11	National Income	380	
12			

• =380